

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0893

Pineville 345 kV
October 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0893. The Interconnection Customer proposes a 773.4 MW large load (data center) facility interconnecting at the Pineville 345 kV substation in the Maryland area. The request was received on November 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 402 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,737,000.

3. Network Impacts

3.1 Load Deliverability Analysis

CLAWSON – ANTIETAM 138 kV line: 152.0% loading (403 MVA) following the outage of the W WAYNESBORO – PINEVILLE SUBSTATION 345 kV; DFAX = 32.5%.

For the loss of W WAYNESBORO SUBSTATION – PINEVILLE 345 kV, the REID – CLAWSON 138 kV loads to 146.2% of its 444 MVA rating. Project DFAX: 32.5%.

REID – PALOGEN 138 kV: 143.6% loading (463 MVA) following the outage of the PINEVILLE – WEST WAYNESBORO 345 kV; DFAX = 32.5%.

For the loss of PINEVILLE – W WAYNESBORO 345 kV, the HIGHWAY – PALOGEN 138 kV loads to 139.2% of its 495 MVA rating. Project DFAX: 32.5%.

The W WAYNESBORO – HIGHWAY 138 kV line is overloaded to 133.5% of its 527 MVA emergency rating for the loss of the PINEVILLE – W WAYNESBORO 345 kV L. The project contributes 32.5% DFAX (251.2 MW) to this overload.

ESS WESEL – PINEVILLE 138 kV: 131.2% loading (435 MVA) following the outage of the CHAMBERSBURG – W WAYNESBORO 345 kV; DFAX = 28.3%.

For the loss of W WAYNESBORO – CHAMBERSBURG 345 kV, the CARREL – BUFFALO TUNGSTEN 138 kV loads to 127.7% of its 449 MVA rating. Project DFAX: 13.5%.

The CRESSEY – ESS WESEL 138 kV is overloaded to 127.5% of its 475 MVA emergency rating for the loss of the CHAMBERSBURG – WEST WAYNESBORO 345 kV. The project contributes 28.3% DFAX (219.1 MW) to this overload.

For the loss of W WAYNESBORO – PINEVILLE 345 kV line, the ANTIETAM – PINEVILLE 138 kV L loads to 126.8% of its 456 MVA rating. Project DFAX: 32.5%.

For the loss of W WAYNESBORO – CHAMBERSBURG 345 kV line, the CHEVMAIN – CRESSEY 138 kV line loads to 125.3% of its 447 MVA rating. Project DFAX: 13.5%.

For the loss of W WAYNESBORO – CHAMBERSBURG 345 kV, the CHEVMAIN – BUFFALO TUNGSTEN 138 kV loads to 118.4% of its 479 MVA rating. Project DFAX: 13.5%.

The CHAMBERSBURG – W WAYNESBORO 345 kV is overloaded to 115.8% of its 1,314 MVA emergency rating for the loss of the CHAMBERSBURG 345/138 kV XFMR. The project contributes 57.0% DFAX (440.7 MW) to this overload.

M105 – WYOMING 138 kV: 100.8% loading (610 MVA) following the outage of the CHAMBERSBURG – WEST WAYNESBORO 345 kV; DFAX = 17.0%.

The M105 – OTSEGO 138 kV L is overloaded to 100.8% of its 599 MVA emergency rating for the loss of the W WAYNESBORO – CHAMBERSBURG 345 kV L. The project contributes 17.0% DFAX (131.3 MW) to this overload.

The N EWISTON – WYOMING 138 kV Line is overloaded to 100.4% of its 629 MVA emergency rating for the loss of the WEST WAYNESBORO – CHAMBERSBURG 345 kV. The project contributes 17.0% DFAX (131.3 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ANTIETAM – CLAWSON 138 kV	Rebuild line with higher-capacity conductor	\$3,274,000	59% (\$1,934,000)
REID – CLAWSON 138 kV	Rebuild line with higher-capacity conductor	\$5,988,000	59% (\$3,537,000)
PALOGEN SUBSTATION – REID 138 kV	Rebuild line with higher-capacity conductor	\$1,012,000	59% (\$598,000)
HIGHWAY – PALOGEN 138 kV Line	Rebuild line with higher-capacity conductor	\$3,149,000	59% (\$1,860,000)
W WAYNESBORO – HIGHWAY 138 kV line	Rebuild line with higher-capacity conductor	\$1,678,000	59% (\$992,000)
ESS WESEL – PINEVILLE 138 kV	Rebuild line with higher-capacity conductor	\$3,577,000	75% (\$2,684,000)
CARREL SUBSTATION – BUFFALO TUNGSTEN SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$1,365,000	36% (\$495,000)
CRESSEY – ESS WESEL 138 kV	Rebuild line with higher-capacity conductor	\$6,172,000	75% (\$4,632,000)
ANTIETAM – PINEVILLE 138 kV	Rebuild line with higher-capacity conductor	\$4,002,000	59% (\$2,364,000)
CHEVMAIN – CRESSEY 138 kV	Rebuild line with higher-capacity conductor	\$8,948,000	36% (\$3,245,000)
BUFFALO TUNGSTEN – CHEVMAIN 138 kV line	Rebuild line with higher-capacity conductor	\$5,980,000	36% (\$2,169,000)
CHAMBERSBURG – WEST WAYNESBORO 345 kV line	Rebuild line with higher-capacity conductor	\$19,645,000	37% (\$7,186,000)
WYOMING – M105 138 kV	Reconductor line	\$303,000	46% (\$138,000)
OTSEGO – M105 138 kV	Reconductor line	\$1,580,000	46% (\$723,000)
N EWISTON – WYOMING 138 kV Line	Reconductor line	\$394,000	46% (\$180,000)
	Total allocated to project		\$32,737,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.